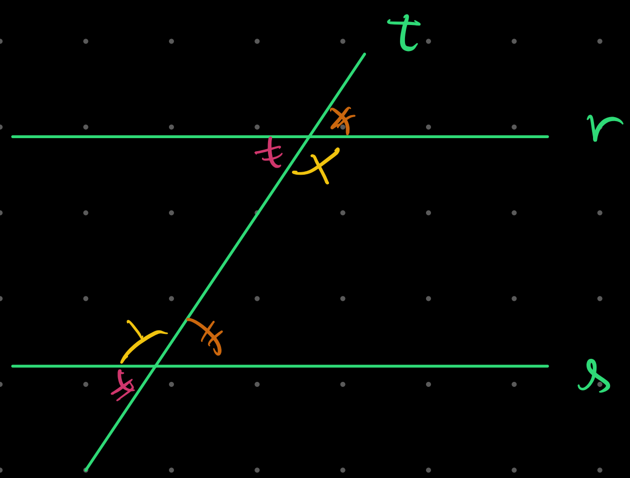
Paralelismo

r || s ↔ r esta no R de s e não se cruzam

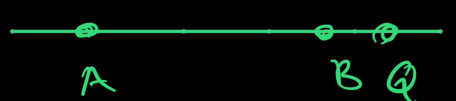


r || s ↔ angulos correspondentes são iguais

r || s ↔ alternos internos iguais

Razão de Segmento

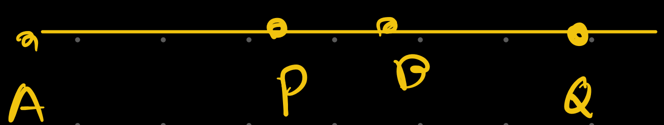
externo



$Q \in \overleftrightarrow{AB} - \overline{AB}$

$$\frac{AQ}{QB} = K$$

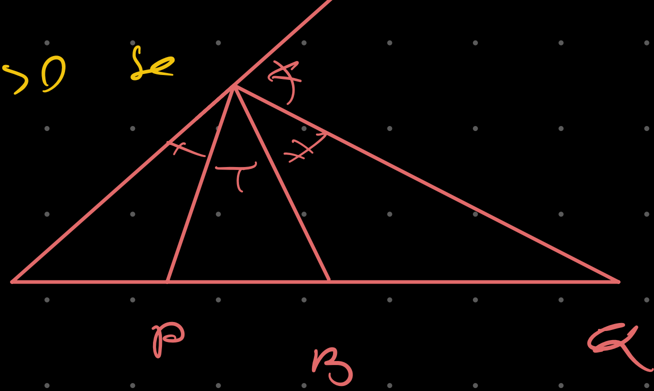
harmonica



P e Q são Divisores Harmônicos

$\overline{AB}$ na razão $K>0$ se

e $\dfrac{AP}{PB} = \dfrac{AQ}{QB} = K$



Desigualdade Triangular

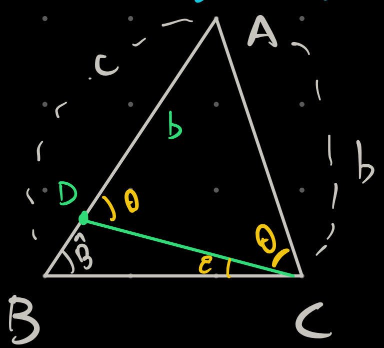
(I) ao maior lado opõe-se o maior ângulo

(II) ao maior ângulo opõe-se o maior lado

(III) $|b-c| < a < b+c$

Demonstração:

(I)



$AB > AC \Rightarrow \hat{C} > \hat{B}$

$D \in AB \mid AD = AC$

$\Downarrow$

$A\hat{D}C = A\hat{C}D = \theta$

$\Downarrow$

$A\hat{D}C = \theta$ é externo de BDC

$\hat{B} + \varepsilon = \theta$ e $\hat{C} = \varepsilon + \theta$

$\hat{C} = \hat{B} + 2\theta > \hat{B}$ c.q.d.

II) Trialidade → redução ao absurdo

$AB > AC$

ou

$AB = AC$

ou

$AB < AC \rightarrow AC > AB$ por (I)

$\hat{B} > \hat{C}$ (X) absurdo

$\rightarrow AB = AC$ pelo teorema do isósceles

$\hat{B} = \hat{C}$ (X) absurdo

Para $\hat{C} > \hat{B} \rightarrow AB > AC$

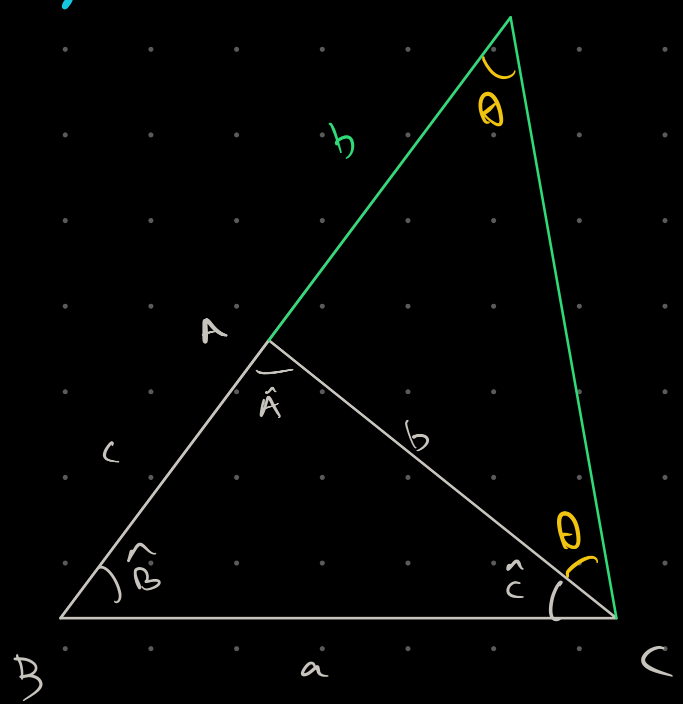
fato tese

Logo $\hat{C} > \hat{B} \rightarrow AB > AC$

então

$\hat{C} > \hat{B} \leftrightarrow AB > AC$

(III)



$B\hat{C}D = \hat{C} + \theta$

$B\hat{D}C = \theta$

$B\hat{C}D > B\hat{D}C$

$BD > BC$

$\underbrace{b + c > a}$

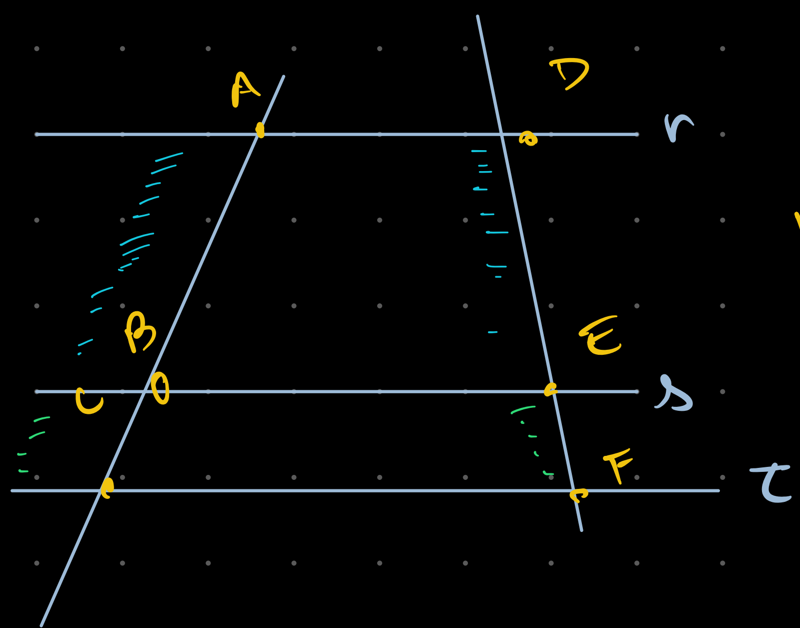
analogamente $b + c > a > |b - c|$

B

O maior lado $a \geq \frac{a+b+c}{3}$

no equilatero $a = \frac{a+b+c}{3}$

Teorema de Tales



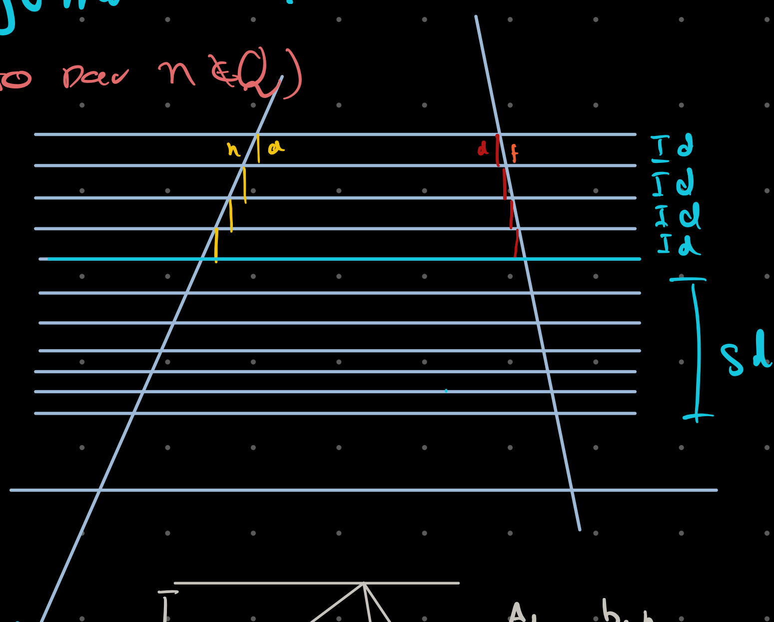
$r // s // t$

razões homólogas são iguais

$$\frac{AB}{BC} = \frac{DE}{EF}$$

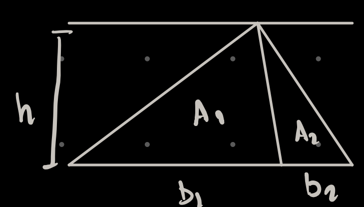
Demo/argumento:

(Incompleto para $n \in \mathbb{Q}$)

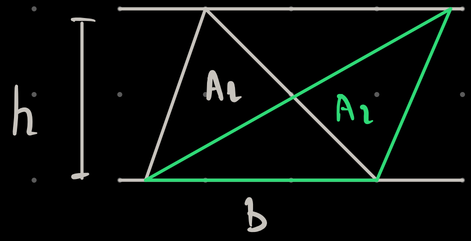


$$\frac{4u}{5u} = \frac{4f}{5f}$$

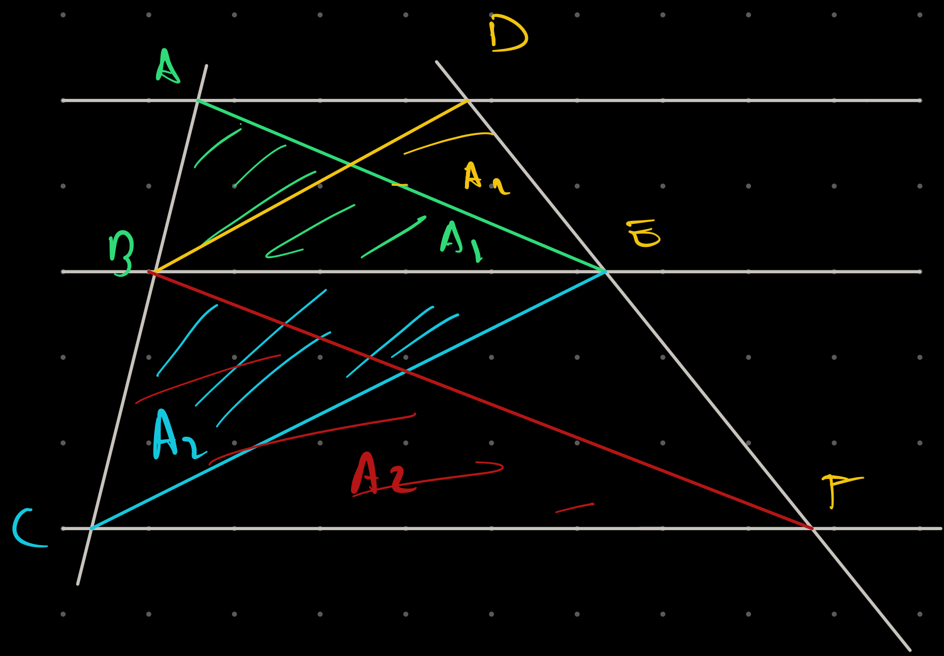
Demo Completa



$$\frac{A_1}{A_2} = \frac{\frac{b_1 h}{2}}{\frac{b_2 h}{2}} = \frac{b_1}{b_2} \Rightarrow$$

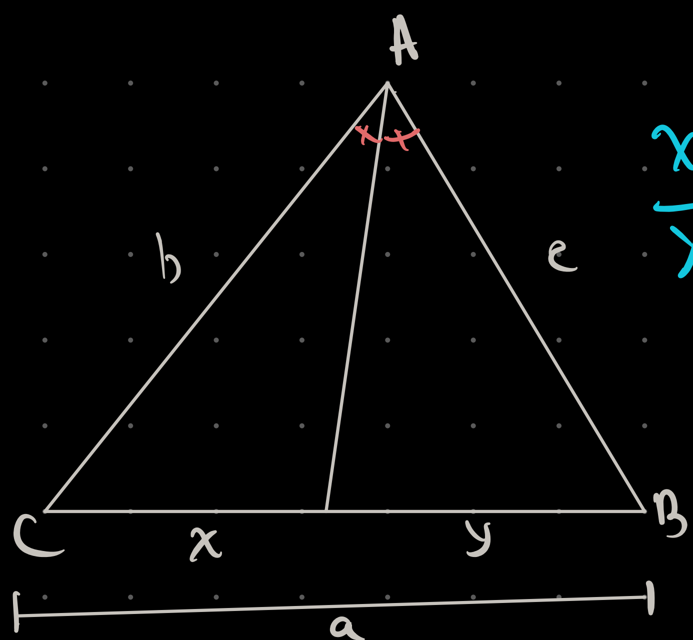


$$A_1 = A_2$$



$$\frac{AB}{BC} = \frac{A_1}{A_2} = \frac{A_1}{A_2} = \frac{DE}{EF} \Rightarrow \frac{AB}{BC} = \frac{DE}{EF}$$

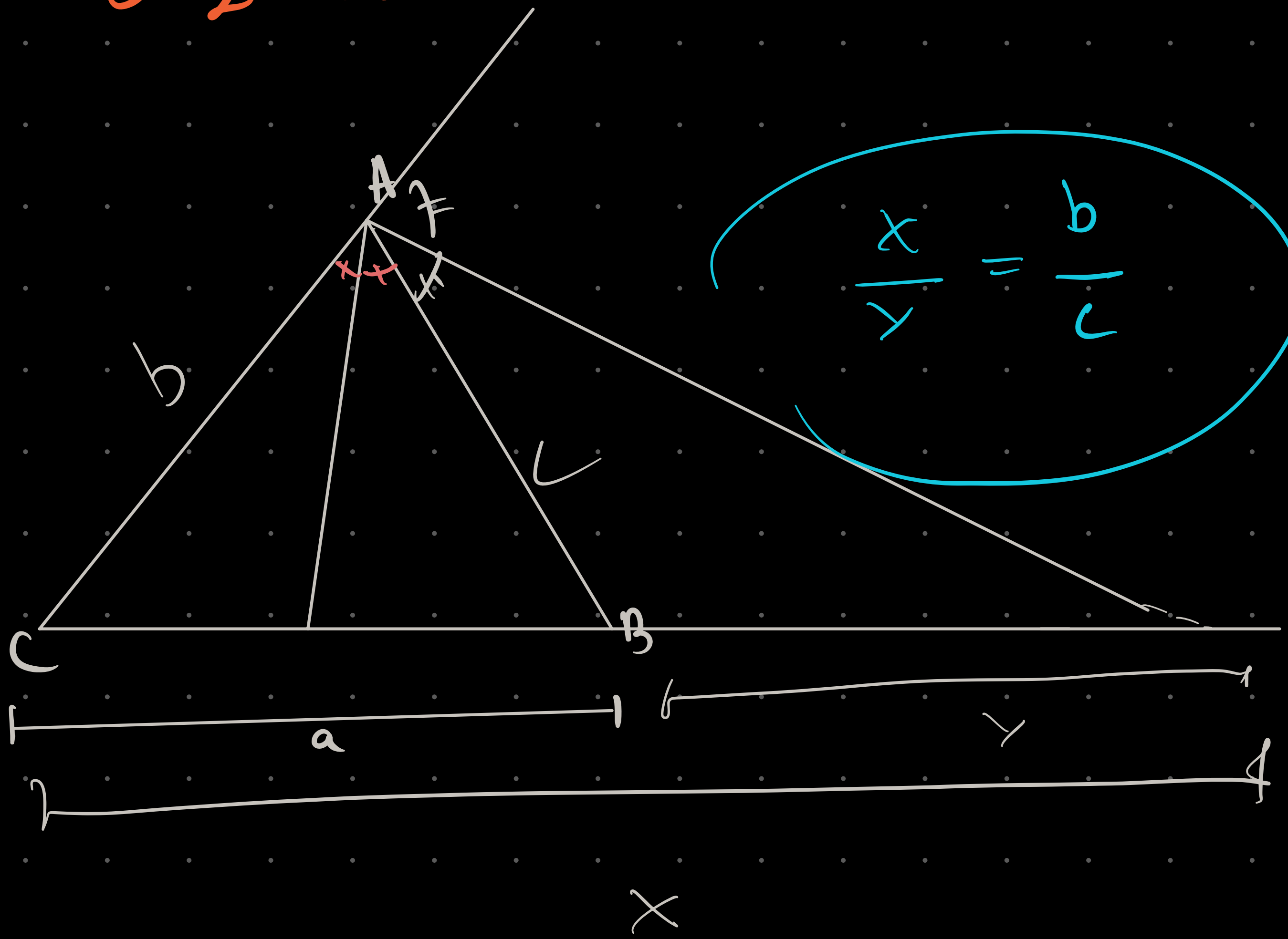
Teorema da BTZ Interna



$$\frac{x}{y} = \frac{b}{c}$$

$$\frac{AI}{ID} = \frac{b+c}{a}$$ corolário

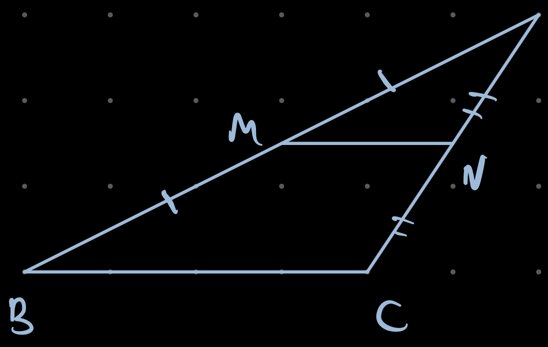
BTZ Externo



$$\frac{x}{y} = \frac{b}{c}$$

Base Média

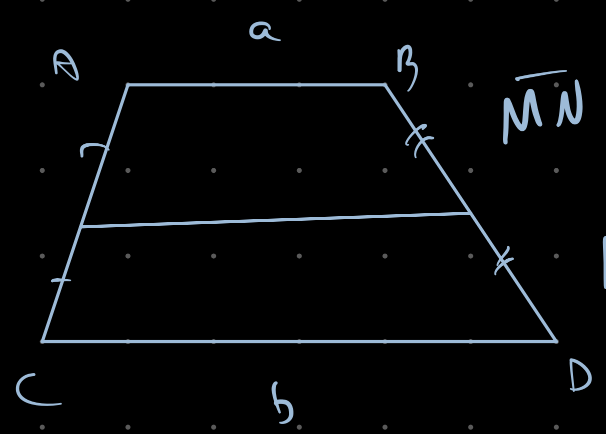
a) Triangulo



$MN \parallel BC$

$MN = \frac{BC}{2}$

b) Trapezio



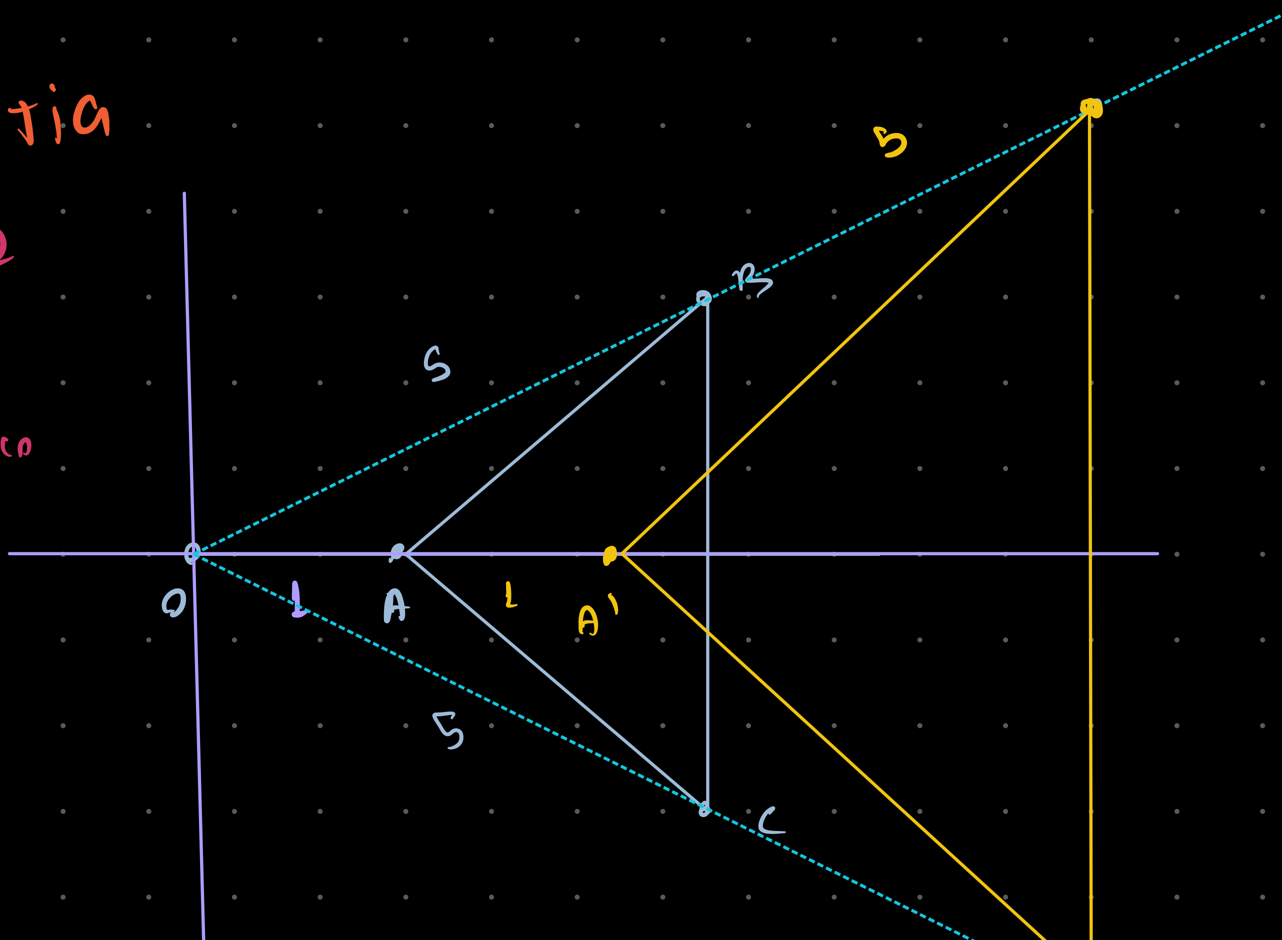
$\overline{MN} \parallel \overline{AB} \parallel CD$

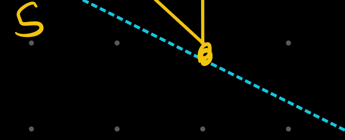
$MN = \frac{a+b}{2}$

Homotetia

razão 2

Centro homotético





$OA' = 2OA$; $OB' = 2OB$; $OC' = 2OC$

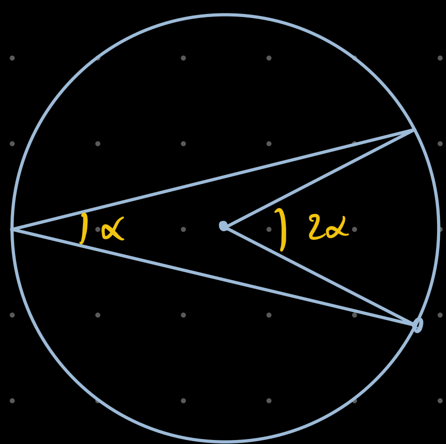
Definição: uma homotetia H de centro O e razão $K \in \mathbb{R}^*$ é uma transformação de pontos do plano (espaço) tal que, dado $P \neq O$:

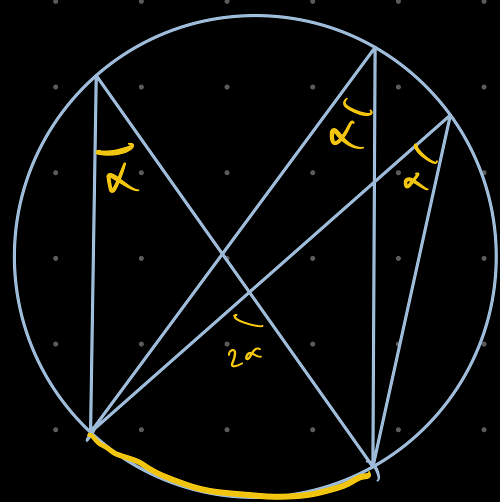
$$H(P) = P' \iff OP' = K\,OP$$

é uma função que recebe um vetor e retorna outro

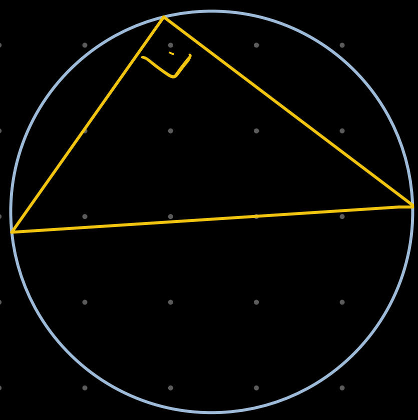
"é uma função linear que age sobre um campo vetorial"

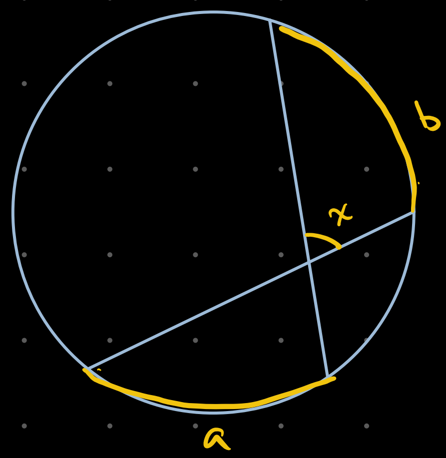
Ângulos na Circunferência





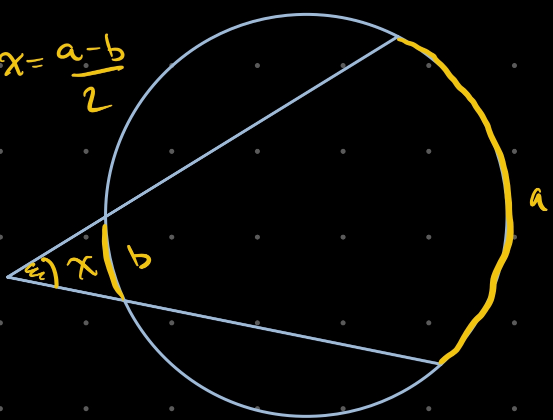
ITA => INSTITUTO De Terapia alternativa

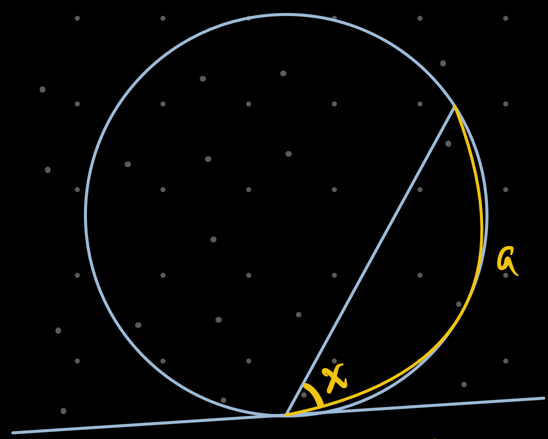




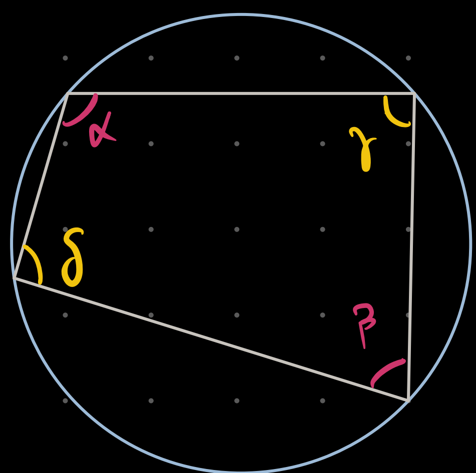
$$x = \frac{a+b}{2}$$

$$x = \frac{a-b}{2}$$



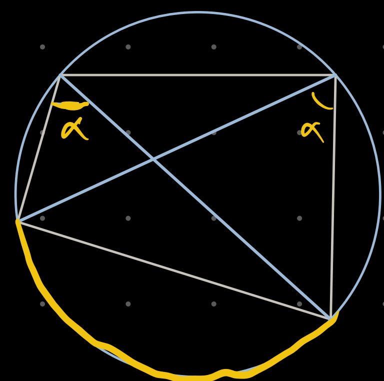


$$x = \frac{a}{2}$$

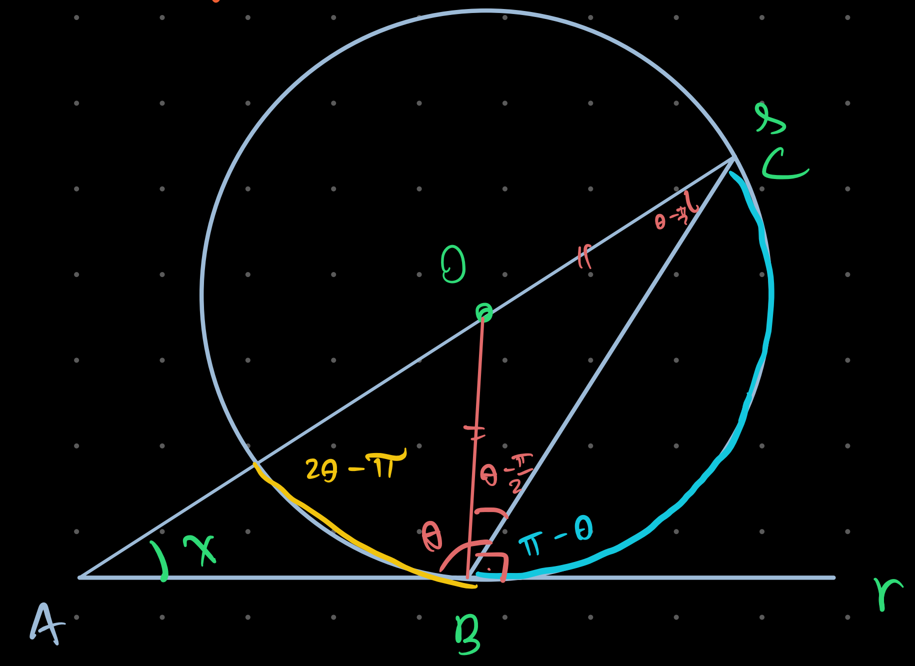


$$\begin{cases} \gamma + \delta = 180^\circ \\ \alpha + \beta = 180^\circ \end{cases}$$

Condição para ser Inscritível



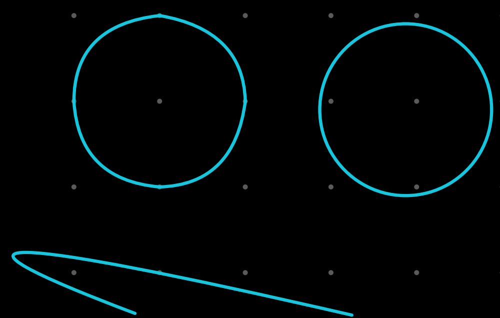
Esercizi

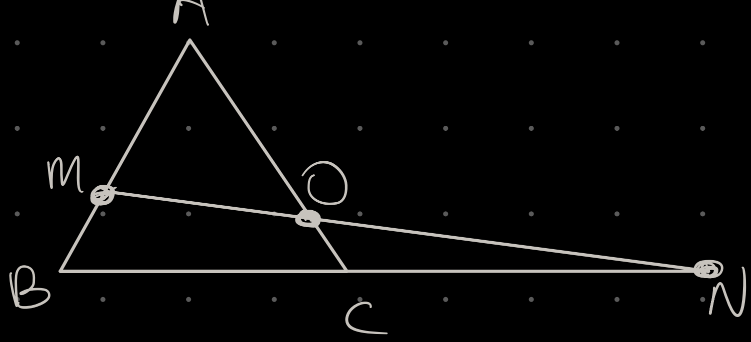


$$x = \frac{2(\pi - \theta) - (2\theta - \pi)}{2} = 2\pi - 2\theta - 2\theta + \pi$$

$$x = \frac{3\pi - 4\theta}{2}$$

$$x = \frac{3\pi}{2} - 2\theta$$

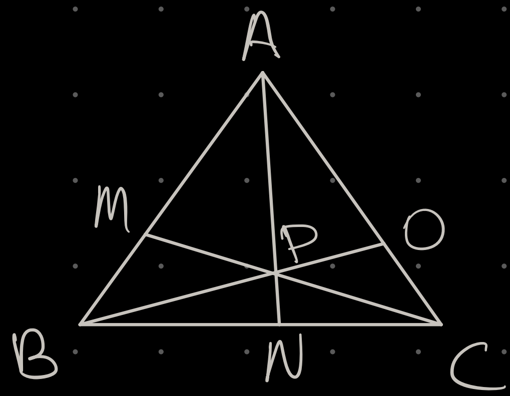




$$\frac{AM}{MB}\cdot\frac{BN}{NC}\cdot\frac{CO}{OA}=1$$

$$\Longleftrightarrow$$

M, N, O colineares



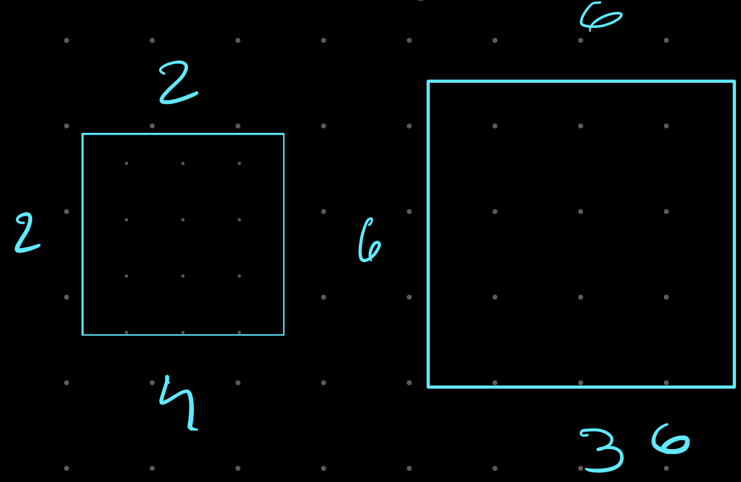
$$\frac{AM}{MB}\cdot\frac{BN}{NC}\cdot\frac{CO}{OA}=1$$

$$\Longleftrightarrow \{P\}$$

Marcão

Razão entre Áreas

1) Entre figuras semelhantes



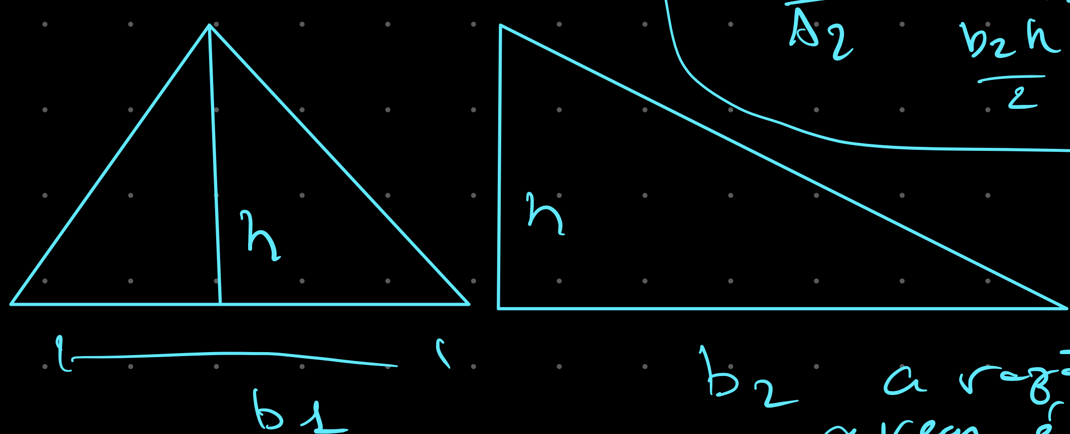
A razão entre áreas é o quadrado da razão dos comprimentos

$$k = \frac{6}{2} = 3$$

$$\frac{A_2}{A_1} = \frac{36}{4} = 9 = k^2$$

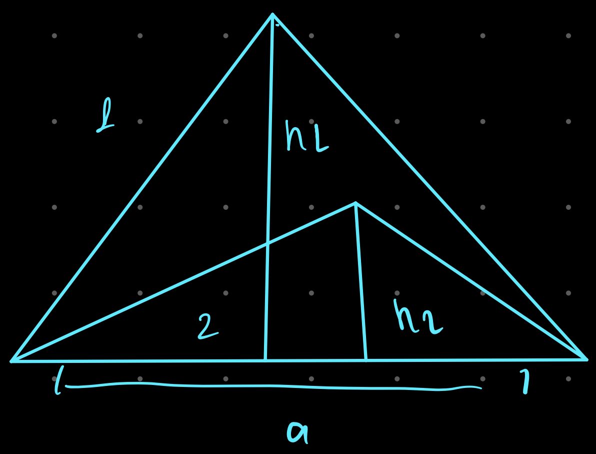
2) Entre triângulos de mesma altura

$$\frac{A_1}{A_2} = \frac{b_1 h/2}{\frac{b_2 h}{2}} = \frac{b_1}{b_2}$$



a razão das áreas é a razão das [illegible]

3) Triangulos de Mesma Base



$$\frac{\frac{a \cdot h_1}{2}}{\frac{a \cdot h_2}{2}} = \frac{h_1}{h_2}$$

razão da area é a razão da altura

GeoEspacial

Volum piramide $\frac{B\,h}{3}$

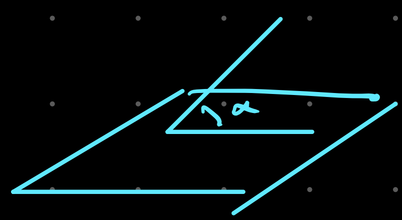
Tetraedro

Area da Superficie $a^2\sqrt{3}$

Altura $\frac{a\sqrt{6}}{3}$

Volume $\frac{a^3\sqrt{2}}{12}$

angulo entre aresta e face



$\operatorname{sen}\alpha = \frac{\sqrt{6}}{3}$

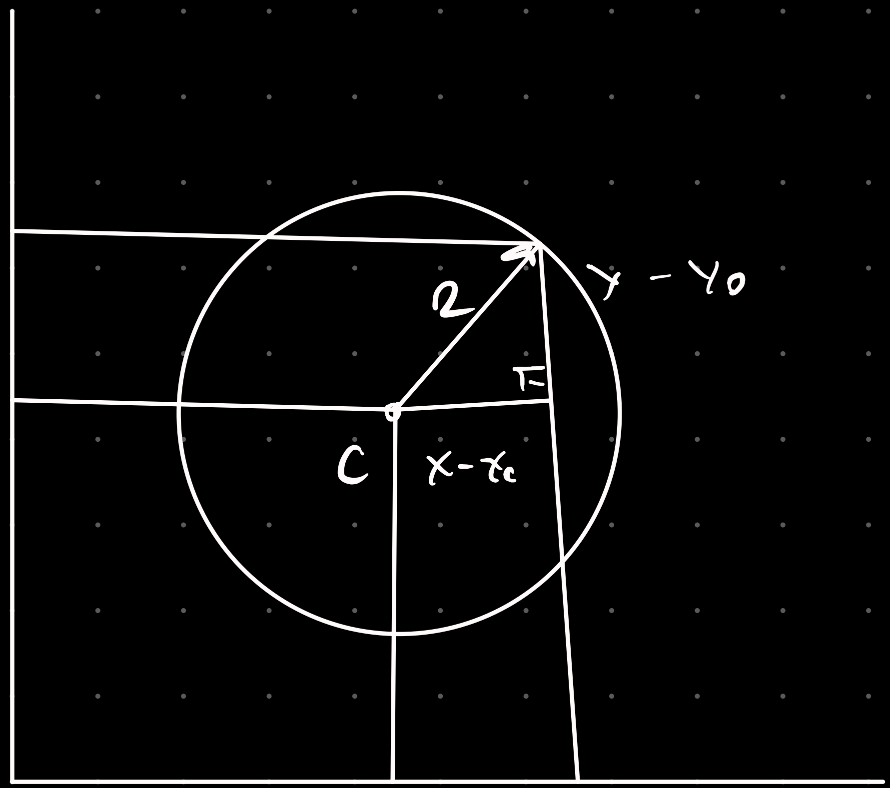
$\cos\alpha = \frac{\sqrt{3}}{3}$

angulo entre Duas faces

$\operatorname{sen}\beta = \frac{2\sqrt{2}}{3}$

$\cos\beta = \frac{1}{3}$

Geo Anal: Equação da Circunferência

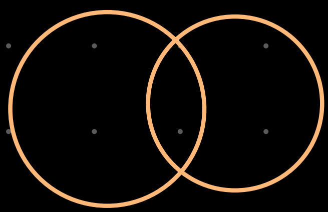


$(x-x_0)^2 + (y-y_0)^2 = R^2$ eq reduzida

$x^2 + y^2 + ax + by + c = 0$ eq Geral

$x_0 = -\frac{a}{2}$ $y_0 = -\frac{b}{2}$ $R = \sqrt{x_0^2 + y_0^2 - c}$

Circunferência Intersecção



$\lambda_1: x^2 + y^2 + ax + by + c = 0$

$\lambda_2: x^2 + y^2 + \alpha x + \beta y + \gamma = 0$

$x^2 + y^2 + ax + by + c = x^2 + y^2 + \alpha x + \beta y + \gamma$

$\longrightarrow$ solu.